Yue Pan

Personal Information

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Education

Princeton University Princeton, NJ
Ph.D. Candidate in Astrophysics (post-generals) *May 2025 – May 2028 (expected)*
M.S. in Astrophysics; Certificate in Computational Science and Engineering *Sep 2023 – May 2025*
Advisor: Prof. Romain Teyssier

University of Chicago Chicago, IL
B.S. in Astrophysics (magna cum laude); GPA: 3.90/4.00 *Sep 2019 – Jun 2023*
Advisor: Prof. Andrey Kravtsov

Research Interests

Computational Astrophysics:

- Developing efficient, physically-motivated models of galaxy formation linking halos, baryons, and environment.
- Modeling reionization feedback by predicting the characteristic mass for halo baryon suppression from local ionization-front arrival time, gas temperature, overdensity, and photoionization rate.
- Modeling stochastic star formation in low-mass galaxies with numerical–statistical methods.
- Building emulators to connect simulation parameters to observable galaxy properties.

High Performance Computing:

- Running and analyzing large cosmological simulations with scalable pipelines.
- Porting, profiling, and optimizing cosmological hydrodynamics codes for NVIDIA GPUs.
- Designing efficient workflows for halo finding, merger trees, and parameter exploration.

Publications

Link to my publications in `NASA ADS`, `arXiv`, and `Google Scholar`

Papers during PhD:
First and second author papers:

1. **Y. Pan**, R. Teyssier, U. P. Steinwandel, A. Pisani. *Much Ado About Nothing: Galaxy Formation and Galactic Outflows in Cosmic Voids*, published in MNRAS in June, 2025. [`arXiv:2503.02938`]
   - Conducted sophisticated simulations with the hydrodynamics code RAMSES in cosmologically underdense environments to unveil the interplay between galaxy formation processes and the large-scale structure. This work required a proficient use of computational models in high-performance computing frameworks.
2. **Y. Pan**, J. Greene, S. Danieli et al. *"The Merian Survey: A Statistical Census of Bright Satellites of Milky Way Analogs"*, published in ApJ in December, 2025.[`arXiv:2512.12846`]

- Lead the analysis of the Merian Survey, the most extensive Milky-Way analog survey, utilizing advanced statistical techniques to characterize the distribution and star-forming activities of satellite galaxies around Milky-Way analogs

3. **Y. Pan**, R. Teyssier et al. *"On-the-Fly Mapping of Intergalactic Metal Enrichment in a Semi-Analytic Galaxy Formation Framework"*, in preparation for submission to MNRAS.
   - Built a mass-conserving 256^3 Eulerian IGM enrichment grid with persistent halo wind bubbles in the `RAMSES` dark-matter-only + `GRUMPY` pipeline; recovered filamentary enrichment at semi-analytic cost, showed that internal enrichment dominates classical-dwarf mass–metallicity relations, and identified two characteristic IGM metallicity peaks.
4. **Y. Pan** et al. *"The Local Filtering Mass of Dwarf-Galaxy Progenitors During Inhomogeneous Reionization"*, in preparation for submission to MNRAS.
   - Using controlled `RAMSES-RT` zoom simulations to predict local filtering mass from ionization-front arrival time, UV intensity, spectral hardness, thermal history, and environment, with baryon retention, cold inflow, and self-shielding as diagnostics.

Collaboration papers:

1. CAMELS Collaboration, including **Y. Pan**. *"The CAMELS Project: Introducing the RAMSES Model"*, in preparation.
   - Contributing to CAMELS-RAMSES, an adaptive-mesh-refinement model for the public CAMELS suite; benchmarking cosmic star-formation history, mass functions, stellar-to-halo mass relations, and matter maps against IllustrisTNG, Simba, Astrid, and Swift-EAGLE while testing responses to cosmology, feedback, and star-formation parameters.
2. A. Mintz, [...], **Y. Pan** et al. *"A non-parametric morphological analysis of Hα emission in bright dwarfs using the first Merian Survey data release"*, published in ApJ in July, 2024. [arXiv:2410.01886].
3. S. Danieli, [...], **Y. Pan** et al. *"Merian: A Wide-Field Imaging Survey of Dwarf Galaxies at $z \sim 0.05 - 0.1$"*, published in ApJ in July, 2024 [arXiv:2410.01884].
4. Y.-F. Luo, [...], **Y. Pan** *"Photometric redshifts for Star-forming Dwarf Galaxies in the Merian Survey"*, in preparation for submission to ApJ.

Papers during undergrad:
First and second author papers:

1. **Y. Pan**, A. Chiti, A. Drlica-Wagner, A. P. Ji, T. S. Li, G. Limberg, D. Tucker, S. Allam. *"Stellar Metallicities from DECam u-band Photometry: A Study of Milky Way Ultra-Faint Dwarf Galaxies"*, published in ApJ in April, 2024. [arXiv:2404.08054]
   - Developed and implemented a novel data processing pipeline utilizing the Dark Energy Survey Camera (DECam) with a custom u-band filter to conduct a comprehensive photometric analysis, identifying metal-poor candidate member stars in three ultra-faint dwarf galaxies. This work required sophisticated data mining techniques and a deep understanding of statistical noise reduction in large datasets.
2. **Y. Pan**, A. Kravtsov, *"Modelling Stochastic Star Formation History of Dwarf Galaxies in GRUMPY"*, submitted to MNRAS in Dec, 2023 [arXiv:2310.08636].
   - Pioneered a stochastic approach to modeling star formation in dwarf galaxies using the semi-analytical model GRUMPY. By integrating advanced statistical methods, including power spectral density analysis, this research provided new insights into the episodic nature of star formation, highlighting my expertise in applying statistical models to complex astrophysical phenomena.
3. **Y. Pan**, C. Simpson, A. Kravtsov, F. A. Gómez, R. J. J. Grand, F. Marinacci, R. Pakmor, V. Manwadkar and C. J. Esmerian, *"Colour and infall time distributions of satellite galaxies in simulated Milky-Way analogs"*, published in MNRAS in Dec, 2022 [arXiv:2208.13805v2].
   - Conducted an extensive analysis of the Auriga simulations to explore the color and infall time distributions of satellite galaxies, utilizing large-scale computational resources. This study showcases my ability to manage and interpret large volumes of complex simulation data, applying my knowledge of galactic modeling to shed light on the various quenching mechanisms that affect dwarf galaxies of different masses.

For other collaboration papers, please see my publication links above.

Conferences, workshops & presentations

1. **New Frontiers in Cosmology (Cosmology under Totality)**
   Invited talk: *The CAMELS Project: Introducing the RAMSES Model*
   August, 2026 — A Coruña, Spain
2. **NVIDIA Open Hackathon at Princeton University**
   GPU computing: `RAMSES` porting, profiling, and optimization with NVIDIA Nsight Compute (`ncu`).
   June 2–4, 2026 — Princeton, NJ
3. **RAMSES User Meeting (RUM) 2026**
   Invited talk: *On-the-Fly Inter-Galactic Medium Enrichment in a Hybrid RAMSES–GRUMPY Galaxy Formation Framework*
   April, 2026 — Seoul, Korea
4. **American Physics Society Global Physics Summit 2026**
   Poster presentation
   March, 2026 — Denver, Colorado
5. **RAMSES User Meeting (RUM) 2024**
   Invited talk: *Much Ado About Nothing: Galaxy Formation in Cosmic Voids*
   April, 2024 — Center for Computational Astrophysics (CCA), Flatiron Institute, NY
6. **Learning the Universe Annual Meeting**
   September, 2023 — CCA, Flatiron Institute, NY
7. **Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond**
   Invited talk: *Stellar Metallicities from DECam u-band Photometry: A Study of Milky Way Ultra-Faint Dwarf Galaxies*
   July, 2023 — CCA, Flatiron Institute, NY
8. **Honors Thesis Presentations**
   Oral presentation: *Modelling Stochastic Star Formation History of Dwarf Galaxies in GRUMPY*
   May, 2023 — Chicago, IL
9. **Wide-Field Spectroscopy vs Galaxy Formation Theory**
   March, 2023 — Biosphere 2, Tucson, AZ
10. **52nd Saas-Fee Advanced Course**, *"The Circum-Galactic Medium across cosmic time : an observational and modeling challenge"*
    March, 2023 — Les Diablerets, Switzerland
11. **241st American Astronomical Society Meeting**
    Invited talk: *"Colour and infall time distributions of satellite galaxies in simulated Milky-Way analogs"*
    January, 2023 — Seattle, WA
12. **DAAD RISE Germany Research Meeting**
    Oral presentation: *"Massive data compression on convergence two-point correlation functions"*
    August, 2022 — Munich, Germany

Computational skills

- Language and software
  - **Proficient in:** Python, LaTeX, Mathematica, Shell/Bash, Git
  - **GPU computing:** NVIDIA GPU porting and optimization; Nsight Compute (`ncu`) profiling
- Computational projects
  - **Experienced with:** the Auriga simulations; RAMSES
  - **Significant contribution to** : the `GRUMPY` semi-analytic galaxy evolution model

Leadership & Teaching experience

- **Department Thunch organizer** — Princeton University, 2023 - 2024
  - Curated and hosted a diverse series of lectures, panels, and discussions on research, career development, astronomy and its social, cultural, and environmental impacts, and EDI-related issues
  - Fed the hungry department members by ordering food from local eateries every Thursday
- **Teaching assistant (TA) of**
  - AST 303 Deciphering the Universe: Research Methods in Astrophysics (Princeton) — Fall, 2024
  - ASTR 20500 Introduction to Python with Astro Statistics (UChicago) — Fall, 2023
  - ASTR 12060 Exoplanets (UChicago) — Summer, 2022
  - ASTR 21100 Computational Astrophysics (UChicago) — Spring, 2022
  - ASTR 21400 Creative Machines and Innovative Instruments (UChicago) — Fall, 2021
- **Society of Physics Students**, *Vice president (2022) & Outreach officer (2021)* — University of Chicago
- **Ryerson Astronomical Society**, *Webmaster (2020-2022)* — University of Chicago

Honors & Awards

American Mensa Member — 2024

Lambda Scholar, *Université de Montréal, Ciela Institute* — 2023

Quad Research Scholar×2, *University of Chicago* — 2021-2023

Dean's Fund for Undergraduate Research Conference ×2, *University of Chicago* — 2022-2023

Odyssey Metcalf Grant, *University of Chicago* — 2022

DAAD RISE Germany Scholar, *The German Academic Exchange Service* — 2022

Dean's List, *University of Chicago* — 2020-2022

Jeff Metcalf Fellowship, *Peking University* — 2020-2021

Community Outreach & Diversity, Equity, and Inclusivity

- **Princeton Local Outreach for High Schoolers** — Dec, 2025
  - Guided local high-school students through hands-on astrophysics activities using Jupyter notebooks.
- **Department Representative for Diversity Talk Series** — July, 2024 – present
  - Identify diversity & equity speakers across physics and astrophysics; invite them to Princeton to speak.
- **Astromatic 2023** — August 2023, Montréal, Canada
  - Helped organize a week-long hackathon at the Mila institute, fostering interdisciplinary collaboration between astrophysics and AI communities to tackle large dataset challenges.
- **STEM Saturdays at Homewood Science Center** — Homewood, IL
  - Engaged primary school students with interactive astrophysics activities, inspiring scientific curiosity by guiding them in creating comets and designing rockets.
- **Science in the Parks: Explore the Night Sky at Big Marsh Park** — Big Marsh Park, IL
  - Guided young learners in astronomical observations, enhancing their understanding of the night sky through hands-on experience with optical telescopes.